

# ELEC5305 Project Proposal

## Project Title

Deep Learning-Based Audio Quality Restoration for Remote Speech Transmission

## Student Information

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## Project Overview

Remote speech transmission is highly susceptible to various sources of degradation, such as background noise, packet loss, echo, and channel distortion. These issues often result in speech signals that are muffled, distorted, or difficult to comprehend. Such degradation not only reduces the quality of everyday online communication but also creates serious challenges for critical applications such as remote education, telemedicine, and collaborative work. Addressing this problem is therefore both practically relevant and academically significant.

The goal of this project is to develop a speech restoration system using modern deep learning techniques. Specifically, the system will focus on recovering degraded speech signals that have been distorted during long-distance transmission. By employing neural network models such as convolutional autoencoders and Transformer-based architectures, the project aims to reconstruct intelligible and perceptually natural speech, providing a listening experience that is closer to the original signal.

## Background and Motivation

Traditional speech enhancement and restoration techniques have relied heavily on classical digital signal processing methods, such as spectral subtraction, Wiener filtering, and statistical noise modeling. While these methods can effectively reduce simple additive noise, they often fail to handle the more complex and nonlinear distortions caused by channel fading, packet loss, and bandwidth constraints. As a result, speech transmitted through unreliable networks remains far from satisfactory in terms of intelligibility and perceptual quality.

Recent advances in deep learning have transformed the field of speech enhancement. Neural networks, particularly convolutional and recurrent models, have demonstrated an impressive ability to learn time–frequency structures and recover missing or corrupted speech components. More recently, Transformer-based architectures have emerged as powerful alternatives for modeling long-term dependencies in audio signals [1]. These innovations provide an exciting opportunity to design models that specifically target the unique challenges of remote speech

transmission, where degradations are not merely additive but also structural in nature.

The motivation for selecting this topic lies in both its academic value and practical impact. From an academic perspective, the project will allow exploration of advanced deep learning techniques in an applied domain, bridging concepts from machine learning, signal processing, and communication systems. From a practical perspective, the outcome of this work could contribute to more robust and reliable online communication systems, offering immediate value to applications such as teleconferencing, distance learning, and healthcare.

## Proposed Methodology

This project will be implemented entirely in MATLAB, which provides a comprehensive set of toolboxes for both signal processing and machine learning. MATLAB will be used to construct the complete workflow, including signal preprocessing, baseline experiments with classical filtering, and the implementation of neural network models. Its integrated environment allows seamless experimentation, visualization, and performance evaluation, making it an appropriate choice for this study.

At the signal processing stage, the Short-Time Fourier Transform (STFT) [2] will be applied to convert time-domain speech signals into spectrograms, capturing the joint temporal and spectral characteristics of speech. A baseline Wiener filtering method will first be implemented to establish a traditional reference system. This comparison will highlight the potential improvements brought by the proposed deep learning methods.

For the machine learning component, the first approach will adopt a convolutional autoencoder, which is effective in learning mappings from degraded to clean spectrograms through hierarchical feature extraction. In later experiments, the study will explore more advanced deep architectures capable of modeling long-term dependencies in audio, aiming to further improve the restoration of speech signals that have undergone severe degradation. Both approaches will be rigorously tested and compared to evaluate their ability to reconstruct intelligible and perceptually clear speech[3].

The training and evaluation of the models will rely on speech datasets collected from publicly available online resources as well as a self-constructed dataset of recorded speech samples. To mimic real-world remote transmission conditions, synthetic degradations such as packet loss, additive noise, channel attenuation, and echo will be introduced into the clean recordings. This will create a diverse dataset that closely represents practical communication scenarios, enabling the proposed methods to be evaluated under realistic conditions.

## Expected Outcomes

The primary expected outcome of this project is a working prototype capable of restoring degraded speech signals to a significantly improved quality level. The prototype will accept distorted remote speech as input and generate an enhanced signal as output.

Performance will be evaluated using both objective and perceptual metrics. Objective metrics include Signal-to-Noise Ratio (SNR) improvement, which measures the degree of noise suppression, and the Perceptual Evaluation of Speech Quality (PESQ) score, which provides a standardized measure of perceived audio quality. In addition, the Short-Time Objective Intelligibility (STOI) index will be employed to quantify improvements in speech intelligibility.

Beyond numerical evaluation, the project will deliver a public GitHub repository containing source code, audio demonstrations, and documentation. This repository will serve as both a technical reference and a demonstration platform, ensuring reproducibility and facilitating further research in this domain.

## Timeline (Weeks 5–13)

The project will be conducted over an eight-week period, divided into four main stages.

Weeks 5–7: Conduct a literature review and collect datasets. Establish baseline experiments using Wiener filtering to provide initial benchmarks.

Weeks 8–9: Implement and test the first deep learning model (convolutional autoencoder). Perform preliminary experiments to verify feasibility.

Weeks 10–11: Focus on optimization and evaluation, including hyperparameter tuning, exploring Transformer-based architectures, and refining training strategies.

Weeks 12–13: Prepare the final project report, complete GitHub documentation, and develop audio demonstrations showcasing the system's effectiveness.

## Conclusion

This project seeks to address a critical challenge in modern communication systems: the degradation of speech quality during remote transmission. By combining traditional signal processing with advanced deep learning architectures, the proposed work aims to restore intelligibility and perceptual quality in degraded speech signals. The expected contributions include not only a working prototype and objective performance improvements but also open-source documentation that can benefit the broader academic and industrial community. Ultimately, the project holds the potential to improve the robustness and accessibility of remote communication technologies in an increasingly connected world.

## Reference

- [1] Yu, Weiwei, et al. "SETransformer: Speech enhancement transformer." *Cognitive Computation* 14.3 (2022): 1152-1158.
- [2] Wang, Zhong-Qiu, et al. "STFT-domain neural speech enhancement with very low algorithmic latency." *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 31 (2022):

397-410.

[3] Oruh, Jane, and Serestina Viriri. "Spectral analysis for automatic speech recognition and enhancement." International Conference on Machine Learning for Networking. Cham: Springer International Publishing, 2020.